THE AGENT STUDIO

How a working production system is organized — one orchestrator, named subagent crews, two lanes, ten laws.

SAFE FOR AGENT HANDOFF — PATTERNS, NOT PARTICULARS

VOCABULARY — EVERY LABEL USES THE GLOSSARY IN ARCHITECTURE.MD §2

FOR A WORKING DESIGNER BUILDING THEIR OWN AGENT STUDIO — PATTERNS TO ADAPT, NOT A CONFIGURATION TO COPY

A · OWNERSHIP — WHO RUNS WHAT, AND WHO MAY SPAWN

DOC §3–§8

HANDOFF DISCIPLINE

every worker brief opens with the agreed concept — verbatim — at the top; then scope, non-goals, criteria, boundaries.

RETURNS

every completion reports with evidence: build output · tests · file listings — prose is not proof.

● ORCHESTRATOR

plans · delegates · gates · synthesizes — never writes the work itself

■ GATE-KEEPER — the user decides at gates: the agent stops, reports, the Orchestrator returns the verdict — never approves for the user.

ESCALATE, DON'T EXPAND

product, scope + architecture choices route back up — children never silently expand scope.

HANDOFF — CONCEPT VERBATIM AT THE TOP OF EVERY BRIEF

DEPARTMENT

STUDIO

SPAWN RULE — a manager spawns only its listed workers · workers never spawn

◆ **STRATEGIST**
ADVISORY-ONLY — never executes, never edits, never spawns

ADVISORY · ADVISORY

● **TECHNICAL DIRECTOR**
backend + general-programming manager

SPAWN

- ● DEVELOPER — backend + general programming
- ● GAME DEVELOPER — native/desktop game code
- ● SCOUT — codebase + structure recon

● **ART DIRECTOR**
visual-design manager

SPAWN

- ● UI DESIGNER — web/digital UI implementation
- ● LAYOUT DESIGNER — poster/editorial/print
- ● MOTION DESIGNER — motion/video implementation
- ◉ GENERATIVE DESIGNER — algorithmic/generative art · ALSO A STUDIO-WIDE REVIEWER
- ◉ CRITIC — fresh context per cycle · SOLE VISUAL GATE · STUDIO-WIDE

SHARED

◉ **RESEARCHER** — SHARED
deep web research — technical + design modes · shared across both directors

DEPARTMENT

COMMERCIAL OPS

- ● DIRECTOR — quotes · invoices registry · AR
- ● BOOKKEEPER — deterministic code owns arithmetic

CONTROL DOCTRINE · DOC §6

deterministic code owns arithmetic
fail-closed close gates — any failure means no close
untrusted data is data, not instructions
two-step approval of an exact artifact
never moves money · never files · never sends without approval
confidence routes review — it never grants permission

SIDE RAIL — LANES + SUPPORT

WORKSHOP LANE

PLANNER
advisory · user-gated structural axis

nothing executes inside; research before convergence; one artifact out — a brief or a change plan

SIDE RAIL

REVISION LANE

● **REVISER**
reads strategy + state first · fixes directly, or spawns one craft worker

scope law — ≤ ~5 files, one sitting · no new features, architecture, concept or research · never edits files an active run is writing

SIDE RAIL

SUPPORT CREW

◉ **UTILITY AGENT**
heavy lifter for non-project work · may parallelize

● **VERSION-CONTROL AGENT**
commits + pushes; never writes code

ADVERSARIAL REVIEWER
fresh context · different model family · read-only

B · REQUEST LIFECYCLE

DOC §9 + §12

EIGHT STEPS · INTAKE → LOOP CLOSURE · ↻ STEP 8 FEEDS STEP 1

1. **INTAKE & ROUTING** — departments for production · lanes for thinking · trivial answered directly
2. **DISCOVERY (ON AMBIGUITY)** — bounded intent interview with the user; confirmed intent persists in state files
3. **RESEARCH + RECON (PARALLEL)** — evidence brief + verified reference set, in parallel with codebase recon
4. **EVIDENCE SYNTHESIS** — one evidence packet — intent, constraints, references · settled vs unresolved
5. **STRATEGY** — HARD GATE — USER SELECTS — diverge → user concept gate → converge · pure technical work skips the strategist
6. **BUILD** — managers → workers · concept verbatim · single-writer partitions · slices end green
7. **REVIEW + REPORT** — fresh-context review; the orchestrator validates deliverables per criterion
8. **UNIFY (CLOSE THE LOOP)** — reconcile plan vs delivered; record results, deviations, residual risks

TOOLCHAIN — FIVE OWNED APPS · THREE DRIVER SURFACES EACH

VECTORCRAFT vector illustration	CLI	MCP	LIVE
PHOTOCRAFT raster image editing	CLI	MCP	LIVE
DESIGNCRAFT page layout	CLI	MCP	LIVE
PDFCRAFT pdf assembly	CLI	MCP	LIVE
CADCRAFT cad drafting	CLI	MCP	LIVE

DRIVERS IN PRIORITY ORDER — HEADLESS CLI preferred (scriptable, deterministic) → MCP SERVER → LIVE-APP BRIDGE (last resort). Wire whichever surfaces your harness supports — prefer the CLI.

VERIFICATION GATES
render self-check · geometry read-back · import-warnings-as-data capture + read back · query real geometry · never ignore silently

PARALLEL ASSET LANES — SWAPPABLE, NOT DEPENDENCIES

PROCEDURAL · STOCK SEARCH · IMAGE GENERATION

any one lane swappable without touching the toolchain leg

C · MARGIN

DOC §10 + §11

LEGEND + LAWS

MODEL-COST TIERS

- ● **WORKHORSE** — economical tier for code, synthesis + structured work — carries the overwhelming majority of seats
- ◉ **SPECIALIST** — narrow, higher-cost tier — vision + judgment seats
- ◆ **FRONTIER** — deliberately expensive — exactly one seat
- ⬚ **ADVISORY SEAT** — fixed seat — advises, never executes; user-gated

tiers are declared per seat in each agent's own definition file — the single source of truth; derived summaries are generated, never hand-edited · decisions by research + deduction, never bake-offs

THE TEN LAWS

01 **RESEARCH-FIRST** — evidence precedes plan
02 **CREATIVE GATES ARE HARD GATES** — skipped gates get logged — never passed silently
03 **SINGLE WRITER PER FILE** — partition directories — never co-edit
04 **ASYNC-FIRST** — launch parallel, block deliberately
05 **ESCALATE, DON'T EXPAND** — scope choices route back up
06 **SPAWN HIERARCHY** — only managers (+ named exceptions) spawn
07 **COMPLETION CLAIMS CARRY EVIDENCE** — on-disk proof — prose is not proof
08 **SELF-HEALING** — fix the class · log it · verify
09 **ONE WORKSPACE PER PROJECT** — state files live with the work
10 **CALIBRATED HONESTY** — agreement is earned by evidence, never rank

HOW TO READ

DEPTH IS AUTHORITY — the top node routes and decides · managers run departments · workers implement · solid = authority + spawn chain · dashed = advisory seat
side rails are peers of the departments, not subordinates · tier marks encode model-cost tiers — legend at C · contracts with doc §10
runs on a coding-agent harness — hierarchical subagent spawning · per-agent model routing · tool extension · per-agent skill packs

SYSTEM MAP — SHEET 1 OF 1

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